

Molecular Crystals

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Manipulating Light-Induced Dynamic Macro-Movement and Static Photonic Properties within 1D Isostructural Hydrogen-Bonded Molecular Cocrystals

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Abstract: Smart molecular crystals with light-driven mechanical responses have received interest owing to their potential uses in molecular machines, artificial muscles, and biomimetics. However, challenges remain in control over both the dynamic photo-mechanical behaviors and static photonic properties of molecular crystals based on the same molecule. Herein, we show the construction of isostructural co-crystals allows their light-induced cracking and jumping behaviors (photosalient effect) to be controlled. Hydrogen-bonded co-crystals from 4-(1-naphthylvinyl)pyridine (NVP) with co-formers (tetrafluoro-4-hydroxybenzoic acid (THA) and tetrafluorobenzoic acid (TA)) crystallize as isostructural crystals, but have different static and dynamic photo-mechanical behaviors. These differences are due to alternations in the orientation of NVP and hydrogen-bonding modes of the co-formers. After light activation, the 1D NVP-TA crystal splits and shears off within 1 s. For NVP-THA, its photostability and high quantum yield give novel photonic properties, including low optical waveguide loss, highly polarized anisotropy, and efficient up-conversion fluorescence.

Introduction

Smart responsive materials that exhibit mechanically dynamic properties upon external stimuli (e.g., heat, pressure, light and solvent atmosphere) have been the focus of broad and intense research owing to their promising applications to actuators, molecular robotics, artificial muscles, and biomimetics.^[1–6] Typically, the fundamental principles behind these dynamic responses are based on the energy conversion and transformation between electric, light, chemical, and/or thermal energies into micro-scale mechanical motion (e.g., rotating, cracking, expansion, bending, folding, and curling).^[7–14] Of the possible natural stimuli, light is advantageous

not only for its wireless actuation and quick responsiveness, but also its ability to adjust irradiation position, intensity, and time. Collectively, these properties allow for tuning of the mechanically responsive level and monitoring the energy conversion process as a whole.^[15,16] The last decade has seen a significant increase in work on photo-mechanical responsive materials, with most systems based on soft matters with well-known photoactive functional groups (e.g., azobenzene, diarylethenes, spiropyrans, anthracene, and chromenes).^[17–20] Usually, basic mechanisms for dynamic mechanical responses are divided into either chemical reactions (e.g., photo-induced dimerization^[21,22]) or changes of the molecular configuration (e.g., *cis-trans* isomerization^[23,24]). For example, Yu and colleagues reported anisotropic and efficient self-healing hydrogels under NIR irradiation and low pH.^[25] Separately, Zhao's group developed a liquid crystal polymer actuator for soft robotics, where UV light application controlled the crosslinking dimerization of anthracene.^[26]

Compared with the hydrogels, liquid crystals, and polymers, dynamic molecular crystals remain in an earlier stage of development. This is likely due to the high crystalline energy and rigidity that has to be overcome during the deformation process.^[27–29] Until recently, photo-mechanical molecular crystals become a topic of keen interest, owing to their utility in future intelligent materials. This is due to the dense and ordered stacking of the crystal structure, which facilitates a rapid energy conversion and coupling between light and mechanical energy; moreover, the deep mechanisms behind this photo-mechanical process are able to be further analyzed based on the well-defined crystal structures.^[30–32] In this sense, co-crystals, referring to crystalline materials that are assembled by two or more molecules into the same crystal lattice in supramolecular architecture and crystal engineering,^[33] have supplied a new platform to design and tune these static photo-functional and photo-mechanical properties. In co-crystals, the molecular stacking fashions and basic intermolecular interactions (e.g., hydrogen bonding, van der Waals and electrostatic forces) can be rationally tailored by selecting suitable target molecule and the co-assembled units (co-formers).^[34–39] Thus, the co-crystals may emerge novel photo-related properties that are different from the sum of individual molecules.^[9,40–51] For example, Morimoto and Ire have developed a co-crystal of a diarylethene compound with perfluoronaphthalene in a 1:2 ratio. The resulting co-crystal had a reversible photo-induced bending motion owing to the photo-generated ring-close reaction.^[52] However, to date, controlling the dynamic photo-mechanical behaviors of the same molecule based on its co-crystal formation remains very

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challenging. This challenge is particularly difficult for isostructural molecular crystals.

Results and Discussion

Herein, we report that the dynamic photo-mechanical and the static photonic properties (including optical waveguide, polarization and up-conversion fluorescence) of molecular crystals are effectively tuned on and off using the design of the isostructural co-crystals. 4-(1-Naphthylvinyl)pyridine (**NVP**, Figure 1) is selected as the target building block owing to its excellent optoelectronic properties; notably, **NVP** has also been applied in logic gates and optical switches.^[53] Moreover, the potential [2+2] cycloaddition of the olefin pairs in **NVP** make it possible to achieve a microscopic photo-mechanical response. Tetrafluoro-4-hydroxybenzoic acid (**THA**) and tetrafluorobenzoic acid (**TA**) both have similar sizes and shapes, and are attractive hydrogen-bond-organized co-former units for supramolecular assemblies, which facilitate the formation of isostructural crystals and modification of their intermolecular interactions and photonic performances. The different numbers of hydrogen-bond sites in **THA** and **TA** may further regulate the process of photo-mechanical conversion. **NVP-TA** has exhibited rapid photo-induced cracking and movement based on the energy conversion from light to mechanical energy, termed the photosalient effect.^[11,12,15,54,55] More specifically, the photosalient effect stands for a mechanical response (including a sudden and rapid burst, forceful jumps, and/or spring off many pieces) after exposure to UV light. This effect presents a visualized process through the release of stress during the photochemical reaction. The isostructural **NVP-THA** has high photostability without light-driven jumping and shows well-defined, static photonic performances in 1D optical waveguide, polarization and up-conversion fluorescence. These findings thus provide new insights into the mechanism behind hydrogen-bonding self-

assembly and static/dynamic photo-responsive performances in isostructural molecular crystalline materials.

Co-crystallized **NVP-THA** and **NVP-TA** were fabricated through a facile, solution-based process in methanol using a 1:1 stoichiometric ratio under ambient conditions. Within one week, this approach yields two types of one-dimensional (1D) needle-like transparent microcrystals. Single-crystal X-ray diffraction (XRD) measurements show the **NVP-THA** and **NVP-TA** crystals both belong to a monoclinic crystal system (space group $P2_1/c$) and have nearly identical cell parameters, presenting with the same two-dimensional (2D) lamellar structures through hydrogen bonding and π - π interactions (Figures S1 and S2). Thus, **NVP-THA** and **NVP-TA** are regarded as isostructural systems. In both co-crystals, proton transfer occurs from the hydrogen-bond donor (**THA** and **TA**) to the acceptor (**NVP**), thus forming a strong charge-assisted hydrogen bond between two assembled units. The connection of **NVP** and **TA/THA** molecules are interacted through hydrogen bonds to form infinite columns (Table S1). Both the **NVP** and co-former units exhibit anti-parallel arrangement modes, in which the **NVP** dimers form an H-type aggregation, with distances of 3.55 and 3.61 Å between the neighboring olefin groups of the adjacent **NVP** (Figure S3). This aggregation facilitates the [2+2] cycloaddition upon UV irradiation. It is also noteworthy that besides the intermolecular hydrogen bonds (e.g., N-H...O, C-H...O and C-H...F) in the **NVP-TA** co-crystal, the **NVP-THA** motif also involves another O-H...O hydrogen bond between **THA** molecules (Figure S4). Additionally, powder XRD profiles of **NVP-THA** and **NVP-TA** are in agreement with their simulated patterns (Figure S5), suggesting the high yield and purity of the as-prepared samples during co-crystallization. The strong (200) and (020) diffraction peaks in the XRD patterns indicate obvious preferred orientations for both **NVP-THA** and **NVP-TA**.

To determine any potential photo-active properties between two types of isostructural molecular crystals, 1D **NVP-TA** and **NVP-THA** crystals were first observed under fluorescence microscopy excited by 365 nm UV light (100 mW cm⁻²). Interestingly, **NVP-TA** crystals exhibit an obvious photo-responsive behavior (e.g., splitting or explosion) upon UV excitation. As shown in Figure 2a, the **NVP-TA** crystal retains its integrity and transparent 1D rod-like structure before irradiation, and has a well-defined green photoemission. After UV light irradiation, the **NVP-TA** crystal starts to crack along the [002] direction obviously, with the fracturing degree increasing with the increasing duration of UV exposure. The green fluorescence at the cracked points also becomes brighter relative to the unbroken locations; this can be a result of the efficient optical waveguiding property in the 1D molecular crystal, which allows fluorescence visualization during the dynamic photo-mechanical conversion process. The fluorescence spectra for **NVP-TA** also confirm that the emission intensity at 490 nm in the same position is increased greatly after splitting (Figure 2b); however, there is no obvious change in the emission wavelength.

It is observed that the photo-responsive cracking time is strongly related to both the crystal size and morphology: when monitoring the **NVP-TA** with a relatively smaller width

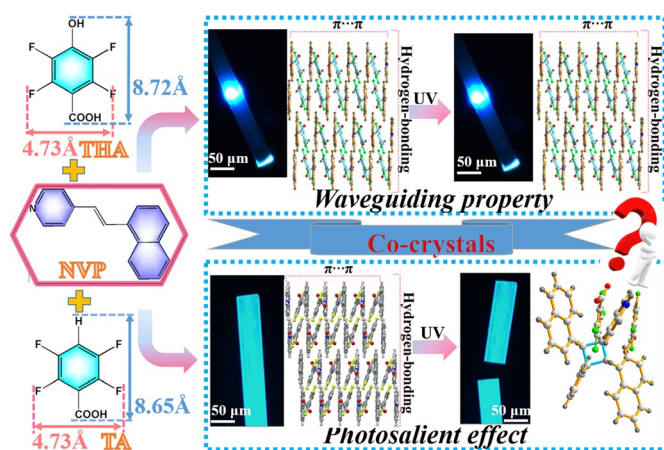


Figure 1. Molecular co-crystals assembled from the chromophore molecule (4-(1-naphthylvinyl) pyridine (**NVP**)) and co-formers (2,3,5,6-tetrafluorobenzoic acid (**TA**), 2,3,5,6-tetrafluoro-4-hydroxybenzoic acid (**THA**)). The two resulting isostructural molecular crystals have notably different photo-responsive behaviors.

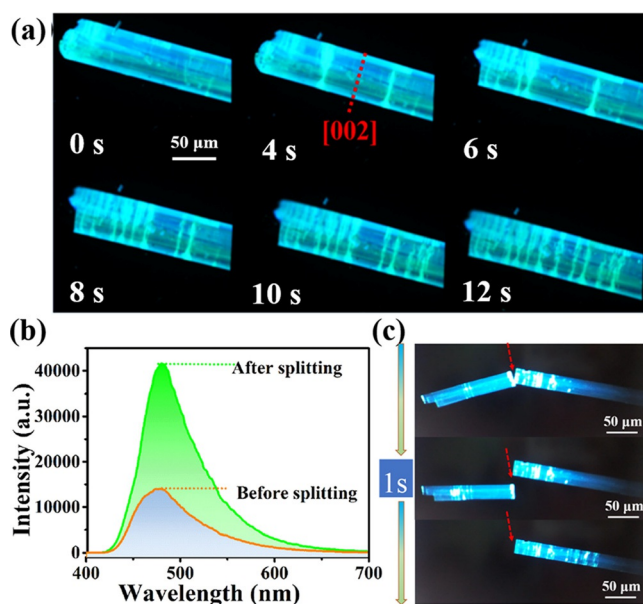


Figure 2. Fluorescence visualization of the light induced splitting and popping processes for the 1D **NVP-TA** crystal (a,c) under 365 nm UV light (100 mW cm^{-2}). b) Fluorescence change for the **NVP-TA** crystal before and after light irradiation.

(approximately $20 \mu\text{m}$), the needle-like crystal splits and shears off rapidly within 1 second (Figure 2c), and the broken section moves out of visible sight (Movies S1–S3). Such observation is understood by considering that the co-crystal's small size allows the photo-mechanical conversion to be achieved more easily. This is due to the short accumulation time needed to overcome the internal crystalline energy of the molecular crystals. Moreover, the bulky **NVP-TA** crystal with a much larger size (approximately 2 cm) is further observed at a macroscopic scale under 365 nm UV light (100 mW cm^{-2}), which undergoes three processes (Movie S4). Firstly, the large crystal begins to flip at 10 s, after which new smaller sections appear when the irradiation time reaches 19 s. Finally, the crystals start to explode into many micro-/nano-sized fragments. The fragments randomly jump away from their original positions in different directions, with the traveling distance of several millimeters as extending irradiation time, which is attributed to the increased stress generated from different axes onto the 1D anisotropic structure. To the best of our understanding, single-component molecular crystals containing a photosalient effect have been continuously reported; however, knowledge regarding the photo-induced free jumping and popping actuation phenomena in the two-component co-crystal systems observed here remains very limited. Since the light-induced dynamic macro-movement and photoemission at the macroscopic level can be easily observed by the naked eye, the **NVP-TA** system described here may serve as a new, easy tool for two-channel UV detection from both spatial movement and fluorescence visualization views.

To better understand the mechanism behind the obvious photo-mechanical characteristics, the **NVP-TA** crystal was subjected to a series of characterizations, including atomic

force microscope (AFM), differential scanning calorimetry (DSC), UV/vis absorption, and PXRD before and after UV light irradiation. Firstly, the microscopic surface morphology of the single crystal was analyzed using AFM (Figure 3). Before photo-irradiation, the surface of the **NVP-TA** crystal is very smooth and flat, with an average roughness value of 1.04 nm . With increasing UV irradiation, the surface gradually becomes rougher, with the roughness increasing to 14.32 nm (for 120 s), which is ascribed to the accumulated strain that is then released during UV irradiation.^[56–58] The DSC results indicate that the sample undergoes a phase transition with the melting point increasing from 155 to 182°C after irradiation (Figure S6). UV/vis absorption shows the maximal absorption in the original crystal occurs at approximately 400 nm , which decreases gradually after UV irradiation, and is accompanied by the blue shift towards 385 nm (Figure S7), illustrating the decrease in degree of π -conjugation. The PXRD results show the disappearance of some diffraction peaks evident in the original crystal after UV irradiation (Figure S8), which tends to be in an amorphous state with a degree of crystalline disruption. These observations indicate that new species and phases appear within the co-crystal during the conversion of UV light into mechanical energy.

To further understand the origin of the photo-mechanical responsive process of **NVP-TA**, we next sought to determine the single-crystal XRD structure from the view a crystallographic study (Figure S3). As described above, before irradiation, the **NVP** molecules in **NVP-TA** are connected in a head-to-tail fashion through π - π interactions, and the distances between the olefin pairs ($3.55 \text{ \AA}$) match the geometric criterion for [2+2] cycloaddition (within the range of $4.20 \text{ \AA}$).^[59,60,28] After a short exposure to UV light, the distance between the $\text{C}=\text{C}$ bonds of **NVP** changes slightly to $3.56 \text{ \AA}$, indicating the structure begins to rearrange in a direction favorable to the cycloaddition reaction. Moreover, the crystallinity of **NVP-TA** becomes poorer, indicated by decreased single-crystal XRD reflection. After further irradiation, the original crystal is almost completely destroyed, rendering it ill-suited for single crystal diffraction. Therefore, we used the recrystallized method to obtain the single crystals of the final photosalient product. Single crystal XRD confirms the occurrence of the photochemical cycloaddition reaction, in which the $\text{C}=\text{C}$ bonds in **NVP** open to form cyclobutane during UV irradiation.

Before the photo-induced dimerization, the **NVP-TA** crystal presents as a monoclinic system with four molecules in each unit cell. The adjacent **NVP** and **TA** molecules are connected by $\text{N}-\text{H}\cdots\text{O}$ ($1.543 \text{ \AA}$ and $2.647 \text{ \AA}$), $\text{C}-\text{H}\cdots\text{F}$ ($2.651 \text{ \AA}$) and $\text{C}-\text{H}\cdots\text{O}$ ($2.422 \text{ \AA}$) hydrogen bonds along the a axis. The **NVP** and **TA** molecules are aligned in parallel, separated by distances of 3.394 and $3.608 \text{ \AA}$, respectively, which stack into 1D columns to overlap along the a axis (Figure S4 and Table S2). For the photoproduct, the newly obtained photochemical dimer compound 1,3-bis(4-pyridyl)-2,4-bis(naphthyl)-cyclobutane (**PNC**) and **TA** are co-crystallized into the triclinic space group ($P\bar{1}$) to form a 2D network along the b axis with six molecules per unit cell. Two neighboring **TA** molecules present a parallel arrangement

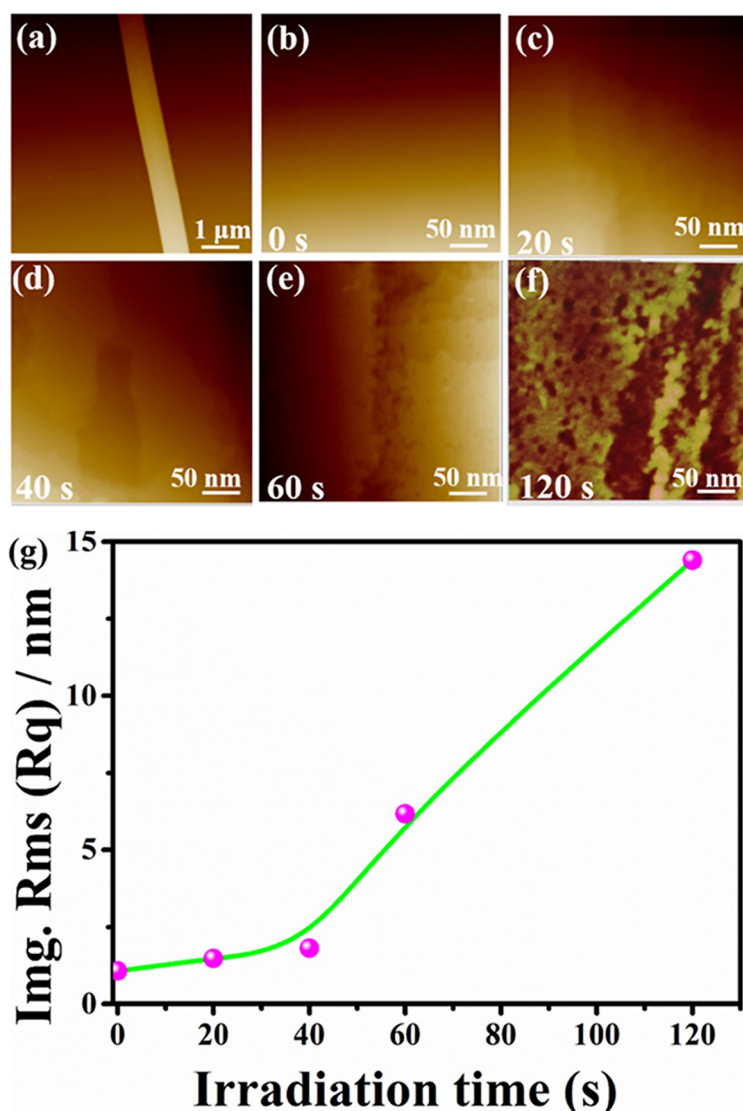


Figure 3. a)–f) Transformation of crystal surface morphology of **NVP-TA** as recorded by AFM with different irradiation times (0 s, 20 s, 40 s, 60 s, and 120 s). g) The surface roughness of **NVP-TA** with different irradiation times.

with a distance of 3.126 Å. **PNC** and **TA** are held together by N–H⋯O (1.684 Å), C–H⋯F (2.465 Å), and C–H⋯O (2.384 Å) hydrogen bonds (Figures S9 and S10). Additionally, the structural parameters for the **PNC** molecule are summarized in Table S3. The photoluminescence quantum yield (PLQY) of **PNC-TA** crystal is 43.61 %, which is higher than the yield before UV irradiation (21.37 %). As observed above, this result is consistent with the enhanced fluorescence intensity at the cracking points. Thus, the molecular arrangements and self-assembly patterns are alternated to a notable degree between the **NVP-TA** and **PNC-TA**.

NMR spectroscopy also provides a direct evidence regarding the photochemical reaction and subsequent structural changes that occurred during photo-mechanical motions (Figure S11–S14). It is noted that new peaks appear at 5.27 ppm and 45.27/43.44 ppm in the ^1H and ^{13}C NMR profiles after UV irradiation, respectively, which are the signals of [2+2] cycloaddition product, **PNC**. The integral values for the cyclobutane ring peaks also increase gradually. Other peaks

remain approximately the same as the initial co-crystal sample along with the time of UV irradiation extending. These results indicate increasing **PNC-TA** content and that the photochemical reaction has not yet been completed given the presence of products and reactants. Based on the integration of the signals in ^1H NMR, the conversion yield within the **NVP-TA** crystal is as high as 80.19 % (Table S4). Given this, the photochemical [2+2] cycloaddition reaction have caused the photo-responsive behaviors for conversion of light into mechanical energy in the **NVP-TA** co-crystal. Moreover, the conversion yield within initial 5 minutes is determined as 20.50 %, which indicates that the threshold value for the percentage of photoproducts is not more than 20.50 % before the jumping occurs. Naumov and co-workers have reported that the 12.5 % photochemical conversion is the lower limit for the occurrence of the photo-mechanical conversion in the single-component molecular crystal.^[56]

Therefore, macro-scaled crystal movement based on the

photosensitive effect requires a portion of the dimer photo-products accumulated in the crystal lattice.

As the **NVP-THA** is the isostructural hydrogen-bonded co-crystal of **NVP-TA**, it is surprising that no similar photo-mechanical behavior is observed (Movie S5). The results indicate that the 1D **NVP-THA** crystal maintains its integrity and crystallinity after irradiation for 360 s. During the photo-irradiation, the fluorescence intensity also remains unchanged (Figure S15). The DSC measurements show nearly identical exothermic peaks positions at 172 °C (Figure S16). Single crystal XRD analysis further confirms that there is no structural change before and after UV irradiation. Thus, the isostructural **NVP-TA** and **NVP-THA** co-crystals present completely different behaviors upon light stimulus, with the former showing rapid light-driven mechanical movement and the latter photo-stability. To understand such obvious discrepancies, detailed structural differences and intermolecular interactions within the co-crystals are further analyzed (Figure 4). As is already known, the [2+2] cycloaddition reactions in the solid state need to fulfill Schmidt's geometric criteria,^[61] i.e., the central distance between the olefin pairs (C=C) should be less than 4.2 Å, and the relative orientation angles (θ_1 , θ_2 and θ_3 , Scheme S1) between the olefin pairs (C=C) prefer their ideal values (0°, 90°, and 90° respectively). In this work, the distances of separation between the olefin pairs in adjacent **NVP** molecules are 3.61 Å and 3.55 Å in the **NVP-THA** and **NVP-TA** co-crystals, respectively, which meet the required distance for the cycloaddition reaction to occur after UV irradiation. However, for the angle criteria (Table S5), the θ_1 , θ_2 and θ_3 angles are 0.31°, 86.43°, and 85.74° respectively in **NVP-TA**, which are much closer to their ideal values than those in **NVP-THA** (0.97°, 78.72°, and 82.70°). Thus, both the distance and angle criteria are beneficial to the photochemical [2+2] cycloaddition reactions for the **NVP-TA** crystal.

On the other hand, for the **NVP-THA** co-crystal, each of the two adjacent **THA** molecules are connected by another strong O—H...O hydrogen bond ($d = 1.758$ Å) with a face-to-face stacking orientation, which is further extended as an infinite 1D zig-zag chain. The Hirshfeld surfaces and two-dimensional (2D) fingerprint are utilized to analyze various intermolecular interaction types for two co-crystals (Figure S17). The red, white, and blue areas on the surfaces represent that the intermolecular contact distances are shorter, equal to and longer than their van der Waals distances, respectively.^[62,63] The contributions of O...H hydrogen bonding in the **NVP-THA** and **NVP-TA** co-crystals can be

confirmed in the Hirshfeld surfaces indicated by the bright red area. It is worth noting that in the bright red area, the color depth for **NVP-THA** is higher than that for **NVP-TA**, which illustrates its stronger hydrogen bonding relative to that in **NVP-TA**. In the 2D fingerprint plots, the F...H interactions cover 28% and 29.5% of the total Hirshfeld surfaces in **NVP-THA** and **NVP-TA**, respectively. The O...H interaction appears as two sharp spikes in the lower left of the plots for the **NVP-THA** co-crystal, which occupies 15.3% of the total Hirshfeld surface with the points in the (di, de) regions of 1.28 Å and 1.00 Å. Obviously, the O...H interaction in the **NVP-THA** co-crystal is stronger compared to the total Hirshfeld surface in the **NVP-TA** (11.4%), indicating the **NVP-THA** co-crystal has close packing and stable lattice energy. This is also consistent with the calculated free volume (Table S6) for the **NVP-TA** crystal (2.1%), which is higher than that of **NVP-THA** (1.5%) and favors effective movement of the molecules in **NVP-TA**.

It has been reported that the photo-reactor molecules will undergo a structural rearrangement process during the [2+2] photodimerization,^[14b] which facilitates the continued generation of the photo-product upon UV irradiation. Herein, an O—H...O hydrogen bond between **THA** appears in **NVP-THA**, which is the only absent interaction in **NVP-TA**. Molecular mechanics calculation shows the hydrogen bonding energy in **NVP-THA** is ca. 2.6 fold than that of the **NVP-TA** (Table S7). This hydrogen bond interaction holds the co-crystallized **NVP** units tightly together in a rigid environment along the (020) direction, and thus restricts and locks the movement and activity of the **NVP** molecules, which creates a constrained environment for further [2+2] cycloaddition between the **NVP**. Therefore, both the geometric criteria for the adjacent **NVP** as well as hydrogen bond effect for co-formers play key roles in designing next-generation photo-mechanical crystalline materials based on 1D isostructural co-crystals.

From a different viewpoint, the high photo-stability of the **NVP-THA** could overcome the well-known photo-bleaching problem present in high-efficiency luminescent materials. Given this, we next sought to investigate the interesting photo-physical and photonic properties of the 1D photo-stable **NVP-THA** co-crystal, including optical waveguide, polarized fluorescence, and up conversion emission. Time-resolved decay emission spectrum of **NVP-THA** reveals the excited-state fluorescence lifetime is 2.963 ns (Figures S18, S19, Table S8), which is higher than that of pure **NVP** (1.573 ns). This indicates that the introduction of **THA** is beneficial to restraining the nonradiative process of **NVP** chromophores based on their intermolecular interactions. Moreover, the PLQY of **NVP-THA** is 33.37% (Table S8), which is higher than most currently reported co-crystals and is helpful to achieve its efficient optical waveguide. Under irradiation with a 365 nm laser, the individual **NVP-THA** co-crystal presents higher blue-green emission intensity on the tips than that in the body of the co-crystal (Figure 5a and b). This is an obvious optical waveguide signal for the 1D microstructure, in which the generated photons are confined in the molecular crystals and can propagate from the main body to the ends. By moving the excitation laser beam, the

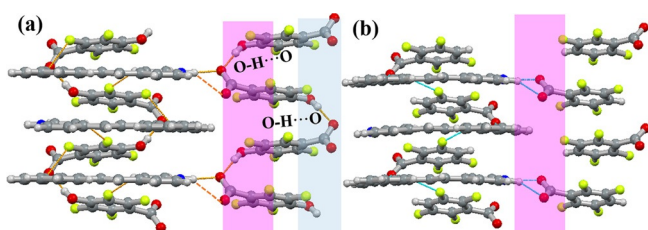


Figure 4. Intermolecular interactions of co-crystals a) **NVP-THA** and b) **NVP-TA**.

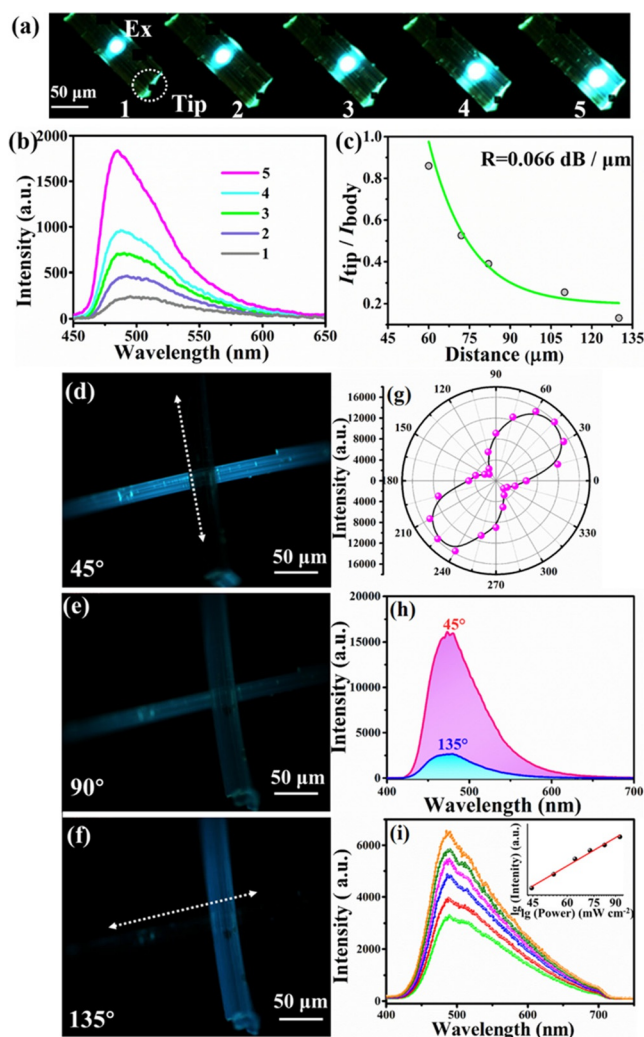


Figure 5. a) The typical fluorescence micrographs images of individual **NVP-THA** under the excitation laser at different spots. b) The PL spectra collected from the tip of **NVP-THA** between the excitation spot and tip as the optical propagation distance increases. c) Fitting an exponential function curve based on $I_{\text{tip}}/I_{\text{body}}$ and distance from the excitation spot to the end to obtain optical-loss coefficient (R) of **NVP-THA**. d)–f) Fluorescence microscope images of **NVP-THA** at different angles (45° , 90° , and 135°). g) Fluorescence intensity of the **NVP-THA** as a function of the polarizer rotation angle; h) Typical polarized fluorescence spectra of **NVP-THA** at different angles (45° and 135°). i) Up-conversion fluorescence spectra of **NVP-THA** excited by 800 nm laser at different pump powers. Inset: Corresponding linear fitting relationship in $\log(\text{intensity})$ with increasing $\log(\text{pump intensity})$ of **NVP-THA**.

optical propagation position is tuned, and the emission intensity and propagation distance of an individual **NVP-THA** can be recorded. The intensity at the tip (I_{tip}) of **NVP-THA** gradually decreases with the excitation distance increasing, wherein the intensity of excitation spot (I_{body}) keeps constant. The intensity ratio $I_{\text{tip}}/I_{\text{body}}$ exhibits a nearly single-exponential decay trend dependent on the propagation distance. The optical-loss coefficient R can be calculated by the function: $I_{\text{tip}}/I_{\text{body}} = A \exp(-RD)$, where D is the distance between the excited spot and the emitting tip. The optical-loss coefficient R of **NVP-THA** is calculated to be $0.066 \text{ dB } \mu\text{m}^{-1}$, which is comparable with current organic molecular crys-

als.^[64–71] Such excellent optical waveguide properties are attributed to the high PLQY and good crystalline quality within the 1D microstructure.

As observed in the crystal structure of **NVP-THA** along a axis and PXRD profile, the **NVP** molecules are highly orientated, indicating the 1D molecular material is extremely anisotropic. After excitation using UV-polarized light at 365 nm, the **NVP-THA** co-crystal exhibits a well-defined polarized emission centered at 480 nm with its maximum (I_{max}) and minimum intensity (I_{min}) located at 45° and 135° , respectively (Figure 5 g and h). It is also observed that the two vertical 1D **NVP-THA** present obviously alternating bright and dark emissions at two typical polarized degrees (Figure 5 d, e and f). This result suggests the rational combination of the two microstructures can serve as an efficient polarized fluorescent switch. The anisotropy value r is calculated as 0.72, based on the formula $r = (I_{\text{max}} - I_{\text{min}})/(I_{\text{max}} + I_{\text{min}})$. To the best of our knowledge, this polarization anisotropy value is higher than that of state-of-the-art 1D organic molecular crystals.^[72] It is expected that this high anisotropy endows the **NVP-THA** co-crystal with high potential for use as a polarized luminescent material.

To probe the photoemission behavior with different excitation modes, we also performed an up-conversion fluorescence test using excitation of the 1D **NVP-THA** co-crystal with a long-wavelength femtosecond pulsed laser (Figure S21). Firstly, the two-photon absorption (TPA) cross section of **NVP-THA** solution ($10^{-5} \text{ mol L}^{-1}$) was detected, and the TPA values are found in the range of 26.9–62.1 GM ($1 \text{ GM} = 10^{-50} \text{ cm}^4 \text{ s photon}^{-1} \text{ molecule}^{-1}$) from 720 to 780 nm (Figures S20). The excitation-dependent up-conversion fluorescence experiment for **NVP-THA** were further conducted with the typical excitation wavelengths (740, 760, 780, 800 and 820 nm). Using excitation with a 740 nm laser light as an example, **NVP-THA** exhibits strong up-conversion fluorescence with maximum emission peaks at approximately 485 nm, which is very close to the position under 365 nm UV light. Due to the centrosymmetric space group, there is a lack of second harmonic generation effect for the co-crystal. Moreover, changing the excitation wavelength ranging from 760 to 820 nm, the shape and position of the emission peak are almost the same as those excited at 740 nm (Figure 5 i and Figure S21). Moreover, the fluorescence intensity exhibits a nearly linear relationship as increasing pump powers, with the corresponding slopes of $\log(\text{intensity})$ to $\log(\text{excitation power})$ as 2.00, 1.92, 1.97, 1.94, 1.86 respectively. This illustrates the two-photon absorption facilitates up-conversion emission, and the photoemission intensity can be finely tuned by controlling the long-wavelength excitation. Therefore, the 1D **NVP-THA** microcrystal can function as a novel blue two-photon light-emitting material for potential non-linear optical applications.

Conclusion

In conclusion, by manipulating the hydrogen-bonding interactions of the isostructural co-crystals, the light-induced dynamic macro-scaled movement is either effectively tuned

or restricted within the 1D molecular co-crystals. The **NVP-THA** exhibits notable photo-induced mechanical popping behavior upon UV irradiation owing to the solid-state [2+2] cycloaddition. This is attributed to the strain created and released during the conversion from light to mechanical energy. The advantages of rapid light-driven movement and the visualization of the obviously changed fluorescence endow the molecular crystal with great potential for the design of high-performance actuators and two-mode UV detection based on crystal engineering. In contrast, no photosensitive effect for the **NVP-THA** co-crystal was observed after irradiation, owing to its deviation from Schmidt's geometric criteria and extra strong O–H...O hydrogen bonds between the **THA** units. Ultimately, these properties restrict the ring addition of ethylene double bonds. Benefitting from their high stability without photo-bleaching, the **NVP-THA** co-crystals present novel micro/nano-sized photonic properties with high-performance polarized fluorescence, two-photon photoemission, and optical waveguide properties, as a result from strong 1D anisotropy and well-defined crystalline quality. Hence, hydrogen-bond organized isostructural co-crystallization provides a promising strategy to tune photo-mechanical and photophysical properties of molecular materials through the selection of suitable building blocks. Using this approach would open up a new way to develop other similar actuators and/or photonic materials towards more effective energy conversion.

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Conflict of interest

The authors declare no conflict of interest.

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